Name ____ Date ___________ Period _____

Domain and Range: All Three Notations

Algebra II Honors | Unit 1 | Day 1 | TEK A2.7.I

Objective: I can write domain and range in verbal, inequality, set, and interval notation, determine a domain algebraically from a function rule, and use union notation for a domain that comes in separate pieces.

Vocabulary

Domain — the set of all _______________ values, the ____-values. **Range** — the set of all _______________ values.

Inequality notation — writes domain or range using _____________________ instead of interval or set notation.

Set-builder notation — { x | condition }, read "the set of all x _______________ the condition holds."

Interval notation — ________ includes an endpoint, ________ excludes it. Infinity always takes a _______________.

Union ∪ — joins two separate intervals when the domain has a __________ in it.

Undefined — an input the rule cannot accept. Two causes in this course: a _____________________ and a _____________________.

Part 1 — From Pairs and Tables

List least to greatest, each value once.

a. { (-4, -6), (0, -1), (2, 4), (3, -7), (7, 7) } D _________________________ R _________________________

b. { (-5, 3), (-1, 0), (2, 3), (6, -8) } D _________________________ R _________________________

c. Why does the range in (b) have three values when there are four pairs? _______________________________

Part 2 — All Three Notations

Complete every cell.

Verbal description	**Inequality**	**Set notation**	**Interval notation**
all real numbers	___________	_______________	___________
all reals at least -6	___________	_______________	___________
all reals less than 5	___________	_______________	___________
all reals from -3 to 3, 3 included only	___________	_______________	___________
all reals except 2	___________	_______________	___________
just the numbers -2, 0, and 4	*(none)*	_______________	___________

The last two rows are the ones to think hardest about. Explain the difference between them:

____

Row 5 is continuous with one point removed; row 6 is ___________________________________

Part 3 — Finding a Domain from the Rule

No graph, no table - reason from the algebra. Ask what the rule forbids, then exclude it.

Function	**What is forbidden?**	**Inequality**	**Interval notation**
f(x) = √(x - 3)	_____________	_________	_____________

$g(x) = 1/(x - 5)$	_______________	___________	_______________
$h(x) = \sqrt{2x + 8}$	_______________	___________	_______________
$k(x) = 1/\sqrt{x - 1}$	_______________	___________	_______________

Look at $k(x)$. Why is the answer $x > 1$ and not $x \geq 1$?

____

At $x = 1$ the radicand is 0, and that would make the denominator _________ — so 1 must be excluded too.

Part 4 — A Domain in Three Pieces

Find the domain of $m(x) = (x + 2)/(x^2 - 9)$. Show the factoring.

____

____

Denominator factors as ________________________, so x cannot be _________________

Domain, interval notation: ____

Domain, set notation: ________________________________

Which notation communicates this domain more efficiently, and why?

____

Part 5 — Going Further

1. Explain why ∞ can never take a bracket, no matter the function.

   ____

2. Write a function whose domain is exactly $[4, \infty)$. ____________________________

3. Write a function whose domain is exactly $(-\infty, 0) \cup (0, \infty)$. __________________________

4. Can a domain be empty? Give a rule with no valid inputs, or argue that none exists.

   ____

   ____

   Sample: ____

Exit Ticket

Give the domain of $f(x) = \sqrt{x + 5}/(x - 2)$ in interval notation, and justify each restriction.

____

____

Domain ________________________________

Two reasons: ____